

 Part 1 – Question Bank**10-Mark Questions**

1. Explain the fundamental principles, characteristics, and essential ingredients of a public-key cryptosystem.
2. Provide a detailed description of the RSA algorithm, including the key generation process and a complete numerical example of encryption and decryption.
3. Discuss the Diffie-Hellman Key Exchange algorithm, explaining its purpose and the step-by-step mathematical process for establishing a shared secret.
4. Describe Elliptic Curve Cryptography (ECC) in detail, highlighting its mathematical foundation and providing a comparison of its performance against RSA.
5. Analyze the security services provided by asymmetric cryptosystems and the specific challenges or threats they face in practical implementation.

5-Mark Questions

6. Compare RSA and ECC based on parameters such as working algorithms, bandwidth savings, and security.
7. Explain the five major ingredients of a public-key encryption scheme.
8. Illustrate the mathematical steps required for RSA Key Generation using prime numbers p and q .
9. Explain the concept of security services (confidentiality, authentication, integrity, and non-repudiation) within asymmetric ciphers.
10. Describe the physical management and size characteristics of keys used in public-key cryptography.
11. Discuss the role of public-key cryptography in online payment services and e-commerce.
12. Explain the process of finding the negative of a point ($-P$) in Elliptic Curve arithmetic.
13. Describe the "Shared Secret" concept in the context of the Diffie-Hellman algorithm.
14. Explain the cubic equation that defines an elliptic curve and its restriction to a finite field.
15. Detail the characteristics of asymmetric algorithms that distinguish them from symmetric ones.

2-Mark Questions

16. Define Plaintext and Ciphertext in a public-key system.
17. Who are the creators of the RSA algorithm and when was it published?
18. State the formula for Euler's Totient Function $\phi(n)$.
19. What is the principal attraction of ECC compared to RSA?
20. Define the term "Primitive Root" as used in Diffie-Hellman.
21. Give the mathematical formulas for RSA Encryption and Decryption.
22. What is the major security issue regarding the substitution of public keys?
23. Contrast the ownership and accessibility of public keys versus private keys.
24. What are the common bit sizes for keys in public-key cryptography?
25. Define "Non-repudiation" in the context of asymmetric algorithms.

EXAM POV NOTES (Q&A FORMAT)**✦ Part 2 – Complete Answer Bank****10-Mark Questions****Q1. Explain the fundamental principles, characteristics, and essential ingredients of a public-key cryptosystem.**

A public-key cryptosystem, or asymmetric algorithm, uses a pair of mathematically related but distinct keys to provide secure communication and identity verification without requiring a pre-shared secret,.

- **Asymmetric Dual-Key Principle:** These algorithms depend on one key for encryption and a distinct but related key for decryption, ensuring that the two processes are logically separated.
- **Computational Infeasibility:** It is essentially impossible to calculate or decide what the decryption key is given only the cryptographic algorithm and the encryption key.
- **Plaintext Component:** This is the readable message or information that is fed into the encryption algorithm as the primary input for the cryptographic process.
- **Encryption Algorithm:** This component performs several mathematical conversions on the plaintext to hide its meaning from unauthorized parties during transmission.
- **Public Key Definition:** This is one part of a key pair that can be shared openly with everyone and is typically used by the sender to encrypt data.
- **Private Key Definition:** This secret key is the matching counterpart to the public key and must be kept strictly secret by the receiver for decryption purposes.
- **Ciphertext Output:** This is the scrambled and unreadable message generated as the output of the encryption algorithm based on the plaintext and the specific key.
- **Decryption Algorithm:** This process takes the ciphertext and the matching private key as input to reconstruct the original, readable plaintext message.
- **Security Services Management:** The system must manage four core services: confidentiality and integrity (via the private key) and authentication and non-repudiation (via the public key).
- **Hardware-Based Maintenance:** Because public-key cryptography keys are very large (512 to 2048 bits), they are maintained in hardware like USB tokens or security modules.

Q2. Provide a detailed description of the RSA algorithm, including the key generation process and the mathematical framework for encryption and decryption.

The RSA algorithm is a foundational public-key cryptosystem named after Rivest, Shamir, and Adleman that relies on the difficulty of prime factorization for its security,.

- **Dual-Key Framework:** RSA works on two different keys where the public key is known to everyone for encryption and the private key is kept secret by the receiver for decryption.
- **Primary Prime Selection:** The first step of key generation involves considering two distinct and very large prime numbers, traditionally denoted as p and q .

- **Modulus Calculation:** The system calculates the public modulus n by multiplying the two chosen prime numbers together ($n = p \times q$).
- **Euler's Totient Function:** The algorithm computes, $\phi(n) = (p - 1)(q - 1)$ which is an essential mathematical step for determining the relationship between exponents.
- **Public Exponent Selection:** A small number e is chosen such that it is co-prime to, $\phi(n)$ satisfying the condition that the greatest common divisor, $\text{GCD}(\phi(n), e) = 1$
- **Private Exponent Calculation:** The user finds the private exponent d such that, $d \times e \text{ mod } \phi(n) = 1$ making d the modular multiplicative inverse of e .
- **Public Key Composition:** The official public key consists of the pair, e, n which is used by anyone wishing to send an encrypted message to the owner.
- **Private Key Composition:** The private key consists of the pair, d, n which must be stored securely to allow for the recovery of original messages.
- **Encryption Mathematical Step:** To convert plaintext P to ciphertext, C the sender applies the modular exponentiation formula. $C = P^e \text{ mod } n$
- **Decryption Mathematical Step:** To retrieve the plaintext P from ciphertext, C the receiver applies the matching formula $P = C^d \text{ mod } n$ to reverse the process.

Q3. Discuss the Diffie-Hellman Key Exchange algorithm, explaining its purpose and the step-by-step mathematical process for establishing a shared secret.

The Diffie-Hellman algorithm is a method used to establish a shared secret key between two parties communicating over an inherently insecure public network.

- **Shared Secret Purpose:** This algorithm is used to generate a common secret key that enables subsequent private communications without the key itself being transmitted.
- **Publicly Available Parameters:** The process begins with two public numbers: a large prime P and its primitive root, G which are known to both parties.
- **Private Value Selection:** Each user independently selects a secret private value, typically denoted as a for Alice and b for Bob.
- **Alice's Public Component:** Alice computes her public value x using the formula $x = G^a \text{ mod } P$ to be sent to Bob.
- **Bob's Public Component:** Bob computes his public value y using the matching formula $y = G^b \text{ mod } P$ to be sent to Alice.
- **Key Exchange Step:** Alice and Bob exchange their generated public values (x and y) across the open network where they could be intercepted.
- **Alice's Secret Derivation:** Alice receives y and computes her version of the shared secret k_a using her private value a via $k_a = y^a \text{ mod } P$
- **Bob's Secret Derivation:** Bob receives x and computes his version of the shared secret k_b using his private value b via $k_b = x^b \text{ mod } P$
- **Algebraic Consistency:** The algorithm ensures that, $k_a = k_b$ as both parties effectively calculate the same mathematical result through different exponentiation sequences.
- **Symmetric Key Utilization:** Once the exchange is complete, both users possess a identical symmetric secret key that can be used for various encryption tasks.

Q4. Describe Elliptic Curve Cryptography (ECC) in detail, highlighting its mathematical foundation and providing a comparison of its performance against RSA.

Elliptic Curve Cryptography (ECC) is a public-key technique that uses the algebraic structure of elliptic curves over finite fields to provide high levels of security.

- **Mathematical Foundation:** ECC is a cryptography technique that works on a mathematical model of elliptic curves defined by specific cubic equations.
- **Finite Field Restriction:** For cryptographic use, the variables and coefficients are restricted to elements in a finite field, resulting in a finite abelian group.
- **Principal Attraction:** Compared to RSA, ECC offers equal security for a much smaller key size, which is the primary reason for its adoption.
- **Processing Efficiency:** By utilizing smaller keys, ECC reduces the processing overhead required for the complex calculations involved in encryption and decryption.
- **Bandwidth Savings:** ECC provides significant bandwidth savings over RSA because the size of the keys and certificates transmitted across networks is greatly reduced.
- **Encryption Speed:** The encryption process in ECC is noted for taking less time than the corresponding process in traditional RSA systems.
- **Decryption Performance:** While overall efficient, the decryption process in ECC typically takes more time than its own encryption process.
- **Modern Security Status:** ECC is considered much safer than RSA in modern contexts and is currently in the process of adapting as a global standard.
- **Key Size Ratios:** As security requirements increase, the size ratio between ECC and RSA keys grows from 1:5 (at low security) to 1:30 (at high security).
- **Future-Proof Security:** At extreme security levels, a 512-bit ECC key is equivalent to a massive 15,360-bit RSA key, demonstrating its long-term viability.

Q5. Analyze the security services provided by asymmetric cryptosystems and the specific challenges or threats they face in practical implementation.

Asymmetric cryptosystems provide a comprehensive security framework while facing significant threats from attackers attempting to impersonate legitimate users.

- **Confidentiality Service:** This service ensures that message content remains private by using the receiver's private key for the decryption of encrypted data.
- **Authentication Service:** Authentication verifies the identity of the user, a process supported specifically by the use of public keys for signature verification.
- **Integrity Service:** This service ensures that the message has not been altered during transit and is considered an essential element of the private-key encryption process.
- **Non-repudiation Service:** Non-repudiation prevents a sender from denying they sent a message and is fundamentally supported by the public-key infrastructure.
- **Masquerading Threat:** A major issue occurs when an attacker masquerades as a legal user to gain unauthorized access to sensitive information.
- **Public Key Substitution:** Attackers may attempt to substitute a legitimate public key with a fake key in a public directory to intercept communications.

- **Connection Interception:** Attackers can intercept the connection between users to capture or alter the keys being exchanged for secure communication.
- **Authenticity Requirements:** Online payment and e-commerce services are only secure when the authenticity of public keys and digital signatures can be guaranteed.
- **Key Size Management:** The large size of keys (512 to 2048 bits) makes them impossible to memorize, requiring specialized hardware for safe storage.
- **Hardware Dependency:** To prevent theft or loss, these complex keys must be maintained in physical devices like USB tokens or hardware security modules.

5-Mark Question Answers

Q6. Compare RSA and ECC based on parameters such as working algorithms, bandwidth savings, and security.

ECC and RSA are the two primary asymmetric standards, differing in their mathematical approach and resource consumption.

Parameters	ECC	RSA
Working Algorithm	Based on a mathematical model of elliptic curves.	Based on the factorization of large prime numbers.
Bandwidth Savings	Gives significant bandwidth savings due to smaller keys.	Low bandwidth savings because keys are much larger.
Encryption Process	The encryption process takes less time.	Generally slower encryption than ECC.
Decryption Process	The decryption process takes more time.	Decryption is relatively efficient in RSA.
Security	Much safer and is the modern adapting standard.	Requires massive keys to maintain modern security.

Q7. Explain the five major ingredients of a public-key encryption scheme.

A public-key encryption scheme is composed of five distinct components that work together to secure data transmission.

- **Plaintext Input:** This is the original readable message or data that is input into the algorithm as the starting input.
- **Encryption Algorithm:** This component performs several mathematical conversions on the plaintext to scramble it into a secure format.
- **Public and Private Keys:** This is a set of two keys selected so that one can be used for encryption and the other for decryption.
- **Ciphertext Output:** This is the final scrambled message generated based on the interaction between the plaintext and the specific key.
- **Decryption Algorithm:** This process accepts the ciphertext and the matching key to recreate the original plaintext for the receiver.

Q8. Illustrate the mathematical steps required for RSA Key Generation.

Generating RSA keys requires a sequence of mathematical operations to ensure the public and private keys are correctly linked.

- **Select Primes:** The process begins by identifying two very large and secret prime numbers, p and q .
- **Calculate Modulus:** The public modulus n is calculated by multiplying the two prime numbers ($n = p \times q$).
- **Compute Totient:** Euler's Totient Function $\phi(n) = (p - 1)(q - 1)$ is calculated to determine the range for exponents.
- **Select Public Exponent:** Choose a value e such that it is co-prime to $\phi(n)$ and falls between 1 and $\phi(n)$.
- **Calculate Private Exponent:** Solve for d using the modular inverse equation $d \times e \bmod \phi(n) = 1$ to finalize the key set.

Q9. Explain the cubic equation that defines an elliptic curve and its restriction to a finite field.

Elliptic curves used in cryptography are defined by specific equations that create the necessary group structure for security.

- **Standard Equation:** An elliptic curve is defined by a cubic equation in two variables with coefficients, typically $y^2 = x^3 + ax + b$.
- **Name Origin:** They are so named because they are described by equations similar to those used for calculating an ellipse's circumference.
- **Field Restriction:** For cryptographic use, all variables and coefficients are strictly restricted to elements within a finite field.
- **Group Definition:** This restriction of values to a finite field results in the mathematical definition of a finite abelian group.
- **Distinction from Ellipses:** Despite their name, elliptic curves are not ellipses, though they share historical mathematical roots in geometry.

Q10. Describe the "Shared Secret" concept in the context of the Diffie-Hellman algorithm.

The shared secret is a unique key generated by two parties without either party actually transmitting the secret itself over the network.

- **Generation Purpose:** The shared secret is established to allow for secret communications while exchanging data over public networks.
- **Public Parameter Use:** It is generated using publicly available numbers P and G alongside private values selected by each user.
- **Key Exchange Mechanism:** Users exchange intermediate public keys which the opposite person uses to generate the final secret key.

- **Mathematical Symmetry:** The process ensures that both parties derive the same secret key ($k_a = k_b$) through their respective calculations.
- **Security Goal:** By the end of the process, users have a symmetric secret key for encryption that was never visible to eavesdroppers.

Q11. Explain the process of finding the negative of a point ($-P$) and the basic rules of point addition in Elliptic Curve arithmetic.

Elliptic curve arithmetic involves geometric and algebraic operations on points that satisfy the curve equation.

- **Point Definition:** A point P on the elliptic curve is represented by a pair of coordinates (x_p, y_p)
- **Finding Negative ($-P$):** The negative of a point P is found at the coordinates $(x_p, -y_p)$ reflecting it across the x-axis.
- **Modular Reduction:** In a finite field mod p the negative coordinate $-y_p$ is calculated as the value that satisfies $(p - y_p) \bmod p$
- **Addition Formula ($P+Q$):** To add two distinct points, the new coordinates (x_R, y_R) are calculated using the slope λ and original coordinates.
- **Lambda (λ) Role:** The value λ represents the slope between points and is calculated differently depending on whether the points are identical or distinct.

Q12. Discuss the role of public-key cryptography in online payment services and e-commerce.

Public-key cryptography is the essential security infrastructure that enables the modern digital economy to function safely,.

- **Private Connection Necessity:** It provides the essential means for users seeking a private connection to exchange encryption keys securely.
- **Identity Verification:** Digital signatures allow e-commerce platforms to sign keys and check the identities of users performing transactions.
- **Payment Service Security:** Online payment services are only ensured when the authenticity of the public key and signature are verified.
- **Transaction Authenticity:** It ensures that the party initiating a payment is the legitimate owner of the account through signature verification.
- **Directory Protection:** It prevents attackers from substituting fake keys in public directories, which could otherwise compromise e-commerce security.

Q13. Detail the physical management and size characteristics of keys used in public-key cryptography.

Due to their complexity, asymmetric keys require specific storage and management protocols to remain secure.

- **Massive Bit Lengths:** Keys generated in public-key cryptography are very large, including lengths of 512, 1024, and 2048 bits.
- **Memorization Difficulty:** These keys are not simple for humans to learn or memorize because of their extreme length and complexity.
- **USB Token Storage:** Because they cannot be memorized, keys are frequently maintained in portable devices such as USB tokens.
- **Security Module Use:** For high-security environments, keys are stored in specialized hardware security modules to prevent unauthorized access.
- **Protection from Alteration:** Proper management is required to ensure attackers do not alter or intercept these keys during storage or use.

Q14. Explain the concept of security services (confidentiality, authentication, integrity, and non-repudiation) as managed by asymmetric ciphers.

Asymmetric cryptosystems distribute the burden of security services across both public and private key components.

- **Confidentiality Service:** This service keeps data secret and is considered an element of the encryption process completed by the private key.
- **Authentication Service:** This verifies user identity and must be supported by the public-key component of the system.
- **Integrity Service:** Integrity ensures data has not been modified and is managed as part of the private-key encryption process.
- **Non-repudiation Service:** This prevents a user from denying a transaction and is specifically supported by the public key infrastructure.
- **Role Specialization:** The asymmetric system is unique because different keys are specialized for different types of security protections.

Q15. Contrast asymmetric algorithms with symmetric ones based on their approach to key distribution and digital signatures.

Asymmetric algorithms were developed to solve specific limitations found in traditional symmetric cryptographic systems.

- **Key Distribution Advantage:** Asymmetric systems provide an essential means of key distribution where users exchange public keys for private connections.
- **Signature Support:** Unlike symmetric systems, public-key cryptography features digital signatures that enable identity verification.
- **Number of Keys:** Symmetric systems use a single shared key, whereas asymmetric systems use two distinct but related keys.
- **Solution to Complexity:** The approach of public-key cryptography derived from an attempt to attack two complex problems of symmetric systems.
- **Public Accessibility:** Asymmetric systems allow one key to be completely public, which is a significant departure from symmetric "secret" models.

2-Mark Question Answers

Q16. Define Plaintext and Ciphertext in a public-key system.

- **Plaintext Definition:** Plaintext is the original readable message or information that is fed into the encryption algorithm as input.
- **Ciphertext Definition:** Ciphertext is the scrambled, unreadable message generated as the output of the encryption process based on the plaintext and key.

Q17. Identify the creators of the RSA algorithm and its year of publication.

- **Algorithm Creators:** The RSA algorithm was created by and named after Ron Rivest, Adi Shamir, and Leonard Adleman.
- **Publication Year:** These three creators officially published the RSA algorithm for public use in the year 1977.

Q18. State the formula for Euler's Totient Function. $\phi(n)$

- **Mathematical Formula:** In the RSA system, the Euler's Totient Function $\phi(n)$ is calculated using the specific formula. $\phi(n) = (p - 1)(q - 1)$
- **Requirement of Variables:** This calculation requires two large secret prime numbers, p and q which are the foundation of the key generation.

Q19. What is the principal attraction of ECC compared to RSA?

- **Security Parity:** The principal attraction is that ECC appears to offer security equal to RSA but with a far smaller key size.
- **Resource Efficiency:** By using these smaller keys, ECC significantly reduces the processing overhead required for cryptographic operations.

Q20. Define the term "Primitive Root" as used in Diffie-Hellman.

- **Variable Definition:** In Diffie-Hellman, the primitive root is represented by the variable, G which is a publicly available number.
- **Role in Exchange:** It is used as the base for modular exponentiation to generate public keys for both parties involved in the exchange.

Q21. Give the mathematical formulas for RSA Encryption and Decryption.

- **Encryption Formula:** RSA encryption is performed using the modular formula, $C = P^e \mod n$ where e is the public exponent.
- **Decryption Formula:** RSA decryption retrieves the message using the matching formula, $P = C^d \mod n$ where d is the private exponent.

Q22. What is the major security issue regarding the substitution of public keys?

- **Masquerading Threat:** An attacker can masquerade as a legal user by substituting a legitimate public key with a fake one in a public directory.
- **Interception Risk:** This substitution allows the attacker to intercept connections or alter keys, effectively compromising the entire secure link.

Q23. Contrast the ownership and accessibility of public keys versus private keys.

- **Public Key Accessibility:** The public key is known to everyone and can be used by any sender to encrypt messages or verify signatures.
- **Private Key Ownership:** The private key must be kept strictly secret and is only owned and used by the intended receiver for decryption.

Q24. What are the common bit sizes for keys in public-key cryptography?

- **Standard Sizes:** Keys generated in public-key cryptography are very large, with common sizes including 512, 1024, and 2048 bits.
- **Storage Reason:** These large sizes are used because they are difficult for attackers to guess, though they require hardware for storage.

Q25. Define "Non-repudiation" in the context of asymmetric algorithms.

- **Service Definition:** Non-repudiation is a security service that prevents a user from denying that they sent a specific message or performed an action.
- **Technological Support:** In asymmetric systems, this service is fundamentally supported by the public-key infrastructure and digital signatures.

Numerical Topics:

1. **RSA Key Generation:** Calculating the modulus ($n=pq$), Euler's Totient ($\phi(n)$), and the modular inverse (d).
2. **RSA Encryption & Decryption:** Performing modular exponentiation
3. $(C=P^{e \bmod n})$ and $P=C^{d \bmod n}$.
4. **Diffie-Hellman Key Exchange:** Calculating public values (x, y) and shared secret keys (k) using modular exponentiation.
5. **Elliptic Curve Arithmetic:** Finding the negative of a point (modular reflection) and point addition (calculating λ, x_R and y_R in a finite field).